

CFB TRENTON, ON, Canada

WMO: 716210

Lat: 44.1178N

Lon: 77.5238W

Elev: 86

StdP: 100.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-18.8	-28.2	0.3	-21.1	-24.8	0.4	-18.0	13.0	-3.3	11.6	-3.5	3.1	340	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	29.3	22.2	27.9	21.4	26.6	20.7	23.6	27.5	22.7	26.4	21.8	25.3	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.2	26.1	21.4	16.3	25.1	20.6	15.4	24.3	71.0	27.5	67.6	26.4	64.3	25.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.0	7.9	-26.3	31.8	3.7	1.9	-29.0	33.2	-31.1	34.4	-33.2	35.5	-35.9	36.9		
			-26.6	25.2	3.6	1.1	-29.2	26.0	-31.3	26.7	-33.4	27.3	-36.0	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-6.7	-5.5	-0.6	6.4	13.0	18.4	21.0	20.2	16.2	9.6	3.2	-2.6	(6)
(7)		DBStd	10.76	6.99	5.90	5.44	4.39	4.04	3.41	2.78	2.80	3.84	4.23	4.66	5.70	(7)
(8)		HDD10.0	2079	518	434	331	122	15	0	0	0	2	60	207	389	(8)
(9)		HDD18.3	4078	777	667	588	357	171	40	7	13	86	271	453	648	(9)
(10)		CDD10.0	1274	0	0	1	15	109	253	340	316	188	48	3	0	(10)
(11)		CDD18.3	233	0	0	0	0	7	43	88	71	22	1	0	0	(11)
(12)		CDH23.3	1525	0	0	0	3	51	279	593	462	134	3	0	0	(12)
(13)		CDH26.7	274	0	0	0	0	6	52	117	79	21	0	0	0	(13)
(14)	Wind	WSAvg	3.5	4.1	4.0	3.9	4.0	3.5	3.1	3.1	2.8	2.9	3.4	3.7	3.8	(14)
(15)	Precipitation	PrecAvg	908	75	61	61	77	78	77	81	69	84	85	77	83	(15)
(16)		PrecMax	1131	127	124	115	145	127	166	228	115	193	205	131	140	(16)
(17)		PrecMin	693	46	25	22	9	24	4	16	22	24	27	12	27	(17)
(18)		PrecStd	114	23	24	29	34	28	42	40	27	36	39	30	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.9	15.7	22.7	27.2	30.3	31.2	30.8	29.0	23.2	16.5	12.3	(19)
(20)			MCWB	8.8	6.4	10.5	15.6	19.9	22.3	23.2	22.4	22.5	18.4	13.1	10.5	(20)
(21)		2%	DB	6.8	6.3	11.6	18.8	24.7	28.2	29.5	28.7	26.8	20.7	14.1	8.8	(21)
(22)			MCWB	5.6	4.4	7.5	11.9	18.1	21.0	22.5	22.0	21.6	17.2	11.8	7.4	(22)
(23)		5%	DB	4.5	4.5	9.1	15.9	22.5	26.4	27.9	27.2	24.7	18.7	12.4	6.9	(23)
(24)			MCWB	3.4	2.9	5.9	10.1	16.5	20.2	21.6	21.2	20.2	15.4	10.2	5.5	(24)
(25)		10%	DB	2.6	2.8	6.8	13.6	20.2	24.8	26.5	25.9	23.0	16.7	10.4	4.9	(25)
(26)			MCWB	1.4	1.4	4.2	8.8	15.0	19.2	20.6	20.6	18.9	13.9	8.3	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	6.8	11.8	16.3	21.2	23.7	25.0	24.8	23.6	19.9	13.9	10.7	(27)
(28)			MCDWB	9.9	7.9	14.7	21.1	25.6	28.3	29.4	28.9	27.6	22.0	15.4	11.7	(28)
(29)		2%	WB	5.6	4.7	8.4	13.3	19.1	22.4	23.7	23.4	22.2	17.6	12.3	7.8	(29)
(30)			MCDWB	6.5	6.2	10.6	17.4	23.3	26.6	27.7	26.8	25.6	19.9	13.9	8.7	(30)
(31)		5%	WB	3.5	3.0	6.2	11.1	17.3	21.2	22.7	22.6	21.0	15.9	10.4	5.7	(31)
(32)			MCDWB	4.3	4.2	8.5	13.7	21.2	25.0	26.4	25.8	24.0	18.1	12.0	6.8	(32)
(33)		10%	WB	1.5	1.6	4.3	9.5	15.8	20.0	21.6	21.6	19.6	14.3	8.5	3.6	(33)
(34)			MCDWB	2.5	2.8	6.7	12.9	19.5	23.6	25.1	24.8	22.3	16.4	10.3	4.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	8.9	8.8	9.7	10.2	9.7	9.6	9.9	10.4	8.9	8.0	7.3	(35)
(36)			MCDBR	8.5	9.1	11.5	13.8	12.7	11.4	10.8	11.2	11.5	11.0	10.7	8.5	(36)
(37)			MCWBR	8.1	8.1	8.4	8.6	7.3	6.3	5.9	6.2	7.0	7.9	9.0	7.8	(37)
(38)		5% WB	MCDWB	8.5	8.7	10.8	12.1	11.4	10.1	9.1	9.2	10.3	10.2	9.8	8.3	(38)
(39)			MCWBR	8.3	8.1	8.5	8.8	7.3	6.2	5.7	5.7	6.7	7.9	8.8	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.306	0.321	0.358	0.381	0.422	0.421	0.410	0.371	0.352	0.311	0.286	(40)	
(41)		taud	2.398	2.362	2.429	2.386	2.371	2.280	2.294	2.314	2.411	2.453	2.517	2.485	(41)	
(42)		Ebn at Noon	828	884	919	906	890	851	847	843	852	822	811	812	(42)	
(43)		Edh at Noon	75	93	100	113	119	131	128	121	101	85	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.33	2.17	3.51	4.51	5.47	5.83	5.94	5.21	4.17	2.46	1.53	1.06	(44)	
(45)		RadStd	0.14	0.21	0.32	0.49	0.54	0.47	0.46	0.28	0.34	0.22	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page